INCH-POUND

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31 January 1990
SUPERSEDING
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19 August 1980

MILITARY SPECIFICATION

COMPRESSOR UNIT, RECIPROCATING,

PWER-DRIVEN, AIR, 15 CFM, 3,500-PSI

This specification is approved for use by all departments and Agencies of the Department of Defense,

1. SCOPE

1.1 Scope This specification covers 15-cfm 3,500-psi, power-driven, portable air compressors.

1.2 Classifications. Compressors shall be of the following types and classes, as specified (see 6.2):

Type I - Skid-mounted.
Type II- Wheel-mounted.

class 1 - Gasline-engine-driven.
class 2 - Electric-motor-driven.
Class 3 - Diesel-engine-driven.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks l The following specifications, standards, and handbooks form a part of thisdocument to the extent specified herein. Unless otherwise specified,the issues of these documetns are those listed in the issue of the Departmeof Defense Idex of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATIN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4310
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SPECIFICATION

FEDERAL

VV-F-800 – Fue Oil, Diesel

MILITARY

MIL-V-173 – varnish, Mositure- and-Fungus-Resistant (for Treatment Of Communications, Electronic, and Associated Equipment).
MIL-P-514 – Plates, identification, Instruction and Marking, Blank.
MIL-T-704 – Treatment and Painting of Material.
MIL-H-3177 – Heaters, Coolant, Engine: Multifuel-Burning 24-Volt D.C.
MIL-H-3199 – inters, Vehicular Compartment, 24 Volt, D.C.
MIL-F-3541 – Fittings, Lubrication.
MIL-C-3600 – Compressors, Rotary, Power-Driven; and Compresors, Reciprocative, Power-Driven Air and Gas (Except Oxygen and Refrigeration), Packaging Of.
MIL-G-3859 – Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-H-4607 – Heater, Engine and Shelter, Ground, Type H-1.
MIL-F-5504 – Filters and Filter Elements, Fluid Pressure, Hydraulic Micronic Type.
MIL-H-7365 – Hose, Air Duct, for Ground Heaters.
MIL-R-8573 – Reservoirs, Air, Nonshatterable Steel.
MIL-G-10924 – Grease, Automotive and Artillery.
MIL-B-11040 – Belt, v: Engine Accessory Drive.
MIL-R-14328 – Rubber Sheet; Synthetic, Medium Soft, General-Purpose Gasket Material (for Extreme Climatic Conditions).
MIL-T-25363 – Tank, Pneumatic Pressure, Aircraft, Glass Fiber.
MIL-C-26058 – Cartridge, Gas Purifier, Extremely Low Dewpoint, Type MA-2 l
MIL-H-26666 – Hose Assembly, Tetrafluoroethylene. Pneumatic High Pressure.
MIL-H-38390 – Hose Assembly, Tetrafluoroethylene, Pneumtic, High Pressure.
MIL-A-46168 – Antifreeze, Ethylene Glycol, INhibited, Heavy Duty, Single Package.
MIL-C-46153 – Coat*, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-H-52471 – Hose and Hose Assemblies, Rubber, Hydraulic Pressure Type, General Specification for.
MIL-H-52471/2 – Hose Assemblies, Rubber, Hydraulic Pressure Type, Type 100R2.
MIL-T-83133 – Turbine Fuel, Aviation Kerosene Type, Grade JP-P8.
MIL-C-83959 – Cylinder, Air Purifier.

STANDARDS

FEDERAL

FED-STD-H28 – Screw-Thread Standards For Federal Services.

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MIL-STD-129 - Marking for shipment and storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-838 - Lubrication of Military Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1300 - Engine, Gasoline; Air-Cooled, 6 BHP, Military Design, Model 4A032 , Installation Procedures.
MIL-STD-1410 - Methods for selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Huron Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MIL-STD-1791 - Air Transport Equipment , General Specification.
MS28889 - Valve, Air, High Pressure Charging.
MS35644 - Filler Neck Assembly, Fuel Tank-Military Vehicles.
MS51331 - Guards, Splash, wheel.
MS51335 - Pintle Assembly, Towing, 18,000 Lbs. Capacity. Manual Release.
MS51338 - Coupler, Drawbar, Ring - Light Duty.
MS52028 - Chassis, Trailer, 3/4 Ton, 2-Wheel, Military Design, M116A1l
MS53063 - Indicator, Air Cleaner, Intake; Restriction Mechanical Type.

HANDBOOK

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives, Preservatives and specialty Products for Use. in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center , (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19 120-5099.)

2.2 Non-Government publications. The following document(s) form a part of thisdocument to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of thedocumetns cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC (ANSI).

B19.1 - Air Compressor System, Safety Standards for.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME

Performance Test Code for Displacement Comprssors, Vacuum Pumps, and Blowers, PTC 9.

Boiler and Pressure Vessel Code, Section VIII, Unfired Pressure Vessels.

Code IX - welding and Brazing Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 439 - Automotive Gasoline.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERCIAN WELDING SOCIETY (AWS)

D1.1 - Structural welding Code, steel*
D1.2 - Structural welding code, Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG-1 - Motors and Generators l

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION

NFPA 70- National Electrical Code (Electrical Wiring and Apparatus) l

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, New York, NY 02110.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC -

National MOTOR Freight Classification

(Appilication for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order section, 2200 Mill Road, Alexandria, VA 22314.)

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RUBBER MANUFACTURERS ASSOCIATION

Engineering Standards for Multiple V-Belt Drive.

(Application for copies should be addresses to the Rubber Manufacturers Association, 1400 K Street, NW, Suite 900, Washington DC 20005).

SOCETY OF AUTOMOTIVE ENGINEERS (SAE)

J321 - Tire Guards for Protection of Operator of Earthmoving Haulage Machines
3492 - Rivets and Riveting.
J682 - Rear Wheel splash and Stone Throw Protection
J683 - Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles

(Application for copies should be addresses to the Sociey of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, E1 Monte, CA 91731.)

TIRE AND RIM ASSOCIATION, INC.

Yearbook

(Application for copies should be addressed to The Tire and Rim Association, Inc., 3200 West Market Street, Suite 304, Akron OH 44313.)

UNIFORM CLASSIFICTION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff f Publishing Officer, Rm 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organization that prepare or distribut the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

REQUIREMENTS

3.1 Description Each compressor unit (herein referred to as compressor) shall consist of the following major components:

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a. **Compressor pump (see 3.9.1).**
b. **Power unit which is either a gasoline engine for class 1 (see 3.12), an electric motor for class 2 (see 3.13), or a diesel engine for class 3 (see 3.12).**
c. **Dehydrator unit (see 3.10.7).**
d. **Moisture separator (see 3.10.6).**
e. **Winterization system, when specified (see 3.17).**
f. **Enclosure (see 3.13 and 3.16.1).**
g. **Mounting base: skid-mounted for type I (see 3.19), or wheel-mounted for type II (see 3.20).**
h. **Coolers, traps, servicing line, and air and oil filters.**
i. **Receiver (see 3.10.10).**

3.2 **First article.** Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The compressor shall be fabricatd from compatible materials, inherently corrosion resistant or treated to provide protection against the various form of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the compressor may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3*3*3 Identification of materials and f fishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent and shall make information available upon request to the contracting officer or designated representative.

3*4 **Recovered materials**. For the purpose of this requirement, recovered materials are thee materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distringuished from virgin raw materials. The components, pieces and parts incorporated in the compressor may be newly fabricated from recovered materials to the maximum extent practicable, provided the compressor produced meet another requirements of this specification. Used, rebuilt or remanufactured components pieces and parts shall not be incorporated in the compressor.

3.5 **Environmental requirements.**

3.5.1 Operation at extreme temperatures. The compressor shall operate as specified herein under the following renditions.

3.5.1.1 Low temperatures The nonwinterized compressor shall start and operate as specified herein for not less than 2 hours at rated capacity in ambient temperature at -25 °F. With a winterization system installed and operating the compressor shall start and operate as specified herein for nut less than 2 hours at rated capacity in ambient temperature -50 'F (see 3.17) .

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3.5.1.2 High temperatures. The compressor, either nonwinterized or winterized, shall start and operate as specified herein at any ambient temperature up to +120 °F for not less than 4 hours at rated capacity.

3.5.2 Rain. The compressor in operating condition shall start and operate as specified herein after exposure to 15 minutes of rainfall at a minimum rate of 3 inches per hour directed at all four sides and top. The compressor shall operate as specified herein for 15 minutes at no load while the rain continues to fall at the minimum rate of 3 inches per hour.

3.6. Safety. All rotating or reciprocating parts and other parts subject to high operation temperatures that are of such a nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating functions. The compressor shall conform to the applicable requirements of ANSI B19.1. Warnings shall be mounted on or near components containing hidden hazards such as automatic blow-down discharges.

3.7 Human factors engineering.

3.7.1 Human factors engineering. The compressor shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. special design emphasis shall be given, but not limite#d to, applicable portions of MIL-STD-1472\ , general requirements(4), control/display integration (5.1), visual displays (5.2),conrols (5.4), labeling (5.5), anthroponetry (5.6), design for maintainability (5.9), and hazards and safety (5.13).

3.7.2 Noise limits. The noise produced by the compressor shall conform to MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.15. The provisions fo MIL-STD-1474 4.3 and 4.4 shall be proviced if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursed and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained form the procuring activity. Hazard sign shall conform to MIL-STD-1474, 4.3, except the sign will stae "HEARING PROTECTION REQUIRED WITHIN (specify) FEET". The sign shall be readable at the distance indicated on the sign. For the purpose of the noise measurement, the operator position is defined as 1.5 feet the compressor.

3.8 Performance requirements.

3.8.1 Rated capacity. Within one hour of operation after starting, the compressor shall deliver not less than 15 cubic feet per minute (cfm) at ASME standard of 14.7 psia, 68 °F ard 36 percent relative humidity at a delivery pressure of 3,500 pounds per square inch (psi) (see 6.6.1 and 6.6.2) when operating under all conditions specified herein. The moisture content of the delivered air shall be no greater than 23.6 parts per million (ppm) by volume (-65 °F dewpoint at 14.7 pounds per square inch absolute (psia)).

3.8.2 Tilted operation. The compressor shall deliver the rated capacity at 3,500 psi for not less than 30 minutes when tilted in any position at an angle of 15 degrees from the horizontal with the oil level in the enqine and compresor on the “low" or “add" mark when checked with compressor unit sitting on a level surface. The compressor shall stand for not less than 4 hours in a

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tilted position without any evidence of leaks; faulty lubrication, cooling, or fuel supply distributionl lubricant spill-over; or reduced performance upon starting after standing in this position.

3.8.3 Endurance. The compressor shall operate as specified herein for 500 hours when tested in accordance with 4.5.2.4. At the end of this 500 hour period, the decrease in output shall be not more than 1.5 cfm when delivering air at a pressure of 3,500 psi.

3.9 Maintainability. The compressor shall operate as specified herein ithout repair or maintenance other than the contractor's recommended normal schedule adjustments and servicing, as established by a maintenance shchedule preparaed and submitted by the contractor prior to test. All major asemblies and installed attachments shall be accessible for maintenance repair and replacement without removal other major assemblies and installed attachments not normally removed. Design for maintainability shall comply with 3.7.1 Cover plates which must be emoved for component adjustment shall be equipped **with quick-disconnect fastenings. Cover plates shall be labeled with** nomenclature for items accessible through then in accordance with MIL-STD-1472. 5.9.9.3 *see 3.7.1). All fasteners shall be fo corrosion-resistance material or shall be tested to be corrosion resistant. All screw threads shall be in accordance with FED-STD-H28. Maximum use shall be made of interdhangeable hardware and fastening devices. Means for drainage of lubricants, fuel and condensate traps shall be in an accessible location and shall drain without splashing on any component or on operating personnel. The drain outlest shall be so located facilitate complete drainage into a suitable oontainer. Each maintenance, assembly or disassembly operation performed as a result of testing shall be accoplished by not more than two men using common tools and specified tools furnished with the compressor.

3.10 Compression systeml

3.10.1 Air compressor pump. The air compressor pump shall be interchangeable as an assembly for both type I and type II compressors and for class, 1 class 2, and class 3 compressors when procured from a single contractor.

3.10.2 Cooling Compresor shall be entirely air cooled. Intercoolers shall be provided to remove the heat of compression between consecutive stages. An aftercooler shall also be provided. All ooolers shall have the capacity to cool the air after compression to temperature of not more than the ambient temperature **+50 °F** up to ambient temperature of **120 °F.** In the ambient temperature range of **+30 to -50 °F,** positive conrrol of cooling shall be provided to prevent valves, traps, piping, and controls from freezing. If a manual device is used, a means shall be provided on the control. Each cooler indicate to the operator the oorrect positioning of the device. Each cooler shall be provided with an adjustable, spring-loaded-type, pressure-relief valve adjusted to operate at the pressure necessary to protect the compressor. On winterized compresors, the comprssor cooling airflow shall be controllable so as to permit efficient generation at an ambient tenperature of **-50 °F.**

3.10.3 Pressure vessels. All pressure vessels shall be rated to withstand 1-1/2 times their normal working pressure and shall conform to the requirements of the ASME Code for Unfired Pressure Vessels unless otherwise specified herein.

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3.10.4 Pressure tubing and fittings. All pressure tubing and fittings shall be rated to withstand 1-1/2 times their normal working pressure and shall be of corrosion-resistant steel containing a minimum of 18 percent chromium and 8 percent nickel. All tubing and fittings shall be installed to minimize and withstand vibration and allow for thermal expansion. Provision shall be made for relieving pressure to the atmosphere from all lines in the system without disassembly of any fitting or component.

3.10.5 Valves. Manually operated shutoff, insolation, and control service valves shall be of the positive shutoff type shall be of corrosion-resistant steel containing a minimum of 18 percent chromium and 8 percent nickel. The valves shall require not more than 60 inch-pounds of torque to open or close under a pressure of 3,5000 psi and over-torquing to 150 inch-pounds shall not damage the valve seats. Valve seats shall be resistant to seat erosion. All valve shall be labeled as to function. Valves shall be provided with double ended arrows that are labeled at each end to indicate functional result (i.e. "OPEN", "CLOSE").

3.10.6 Moisture separator. A mechanical-type moisture separator shall be provided downstream from the aftercooler to remove condensed moisture and oil. The separator shall have a minimum operating pressure of 3,500 psi and a proof pressure of 5,250 psi, and shall be of a corrosion-resistant material. The separator shall have a volumetric capacity large enough to contain the moisture and oil which would accumulate in 20 minutes while the compressor is operating at rated capacity with a relative humidity of not less than 85 percent and an ambient temperature of 120 °F. A valve shall be provided for periodic drainage of the moisture separator. Periodic drainage shall not be required more frequently than 15 minute intervals under all conditions specified herein. In addition, automatic drainage of the moisture separator shall be provided on shutdown of the compressor. Provision shall be made to prevent freezing of moisture when operating at ambient temperatures below 32 °F. When an electrical heater blanket is utilized, it shall be actuated by a manual switch. Provisions shall be made to prevent actuation of the heater when the unit is not is use.

3.10.7 Dehydator. The chemical-type dehydrators shall be provided and shall be of the replaceable cartridge type, conforming to MIL-C-83959 and provided with an MA-2 cartridge confroming to MIL-C-26058. Provisions shall be made for relieving pressure to the atmoshphere.

3.10.7.1 Mounting. The cartridge holder shall be rigidly mounted in an accessible location and in a vertical position. The cartridge holder shall be connected so that the air leaving the moisture separator will enter the first cartridge holder at the bottom, leave at the upper connection, enter the second cartridge holder at the bottom connection, and leave at the top.

3.10.7.2 Operating instructions. Instructions for changing the cartridges in the cartridge holders shall be provided on an instruction plate (see 3.23). Instructions shall also include the required hours between cartridge changes for not less than three temperature ranges to maintain a delivery air dewpoint of -65 °F as determined during preproduction testing. Interpolation and extrapolation of test results will be permitted. A blank space shall be provided on the instruction plate on which the hourmeter reading can be recorded and

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erased. The cartridge holders shall be labeled No. 1 (cartridge holder following moisture separator) and No. 2 (outlet cartridge holder). Change cartirdges as follcws:

a. Discard cartridge in Nol 1 (inlet) holder.
b. Transfer cartridge form Nol 2 (outlet) holder to No. 1 holder.
c. Place fresh cartridge in No. 2 holder.

3.10.7.3 Pressure control valve. An automatic 1/4 inch pressure control valve shall be provided to prevent the passage of air through the moisture separator or dehydrator at pressures less than 2,000 psi. A check valve shall be provided to prevent reverse airflow.

3.10.8 Air filter.

3.10.8.1 Intake air filter. A dry type air filter shall be furnished for air being inducted into the first stage of the air compressor. Each air filter shall be provided with an air indicator conforming to MS53063-3 which shall be either located at the air cleaner or the control panel (see 3.15) in a position visible to the operator.

3.10.8.2 Moisture separator inlet air filter. An accessible and easily serviced filter shall be provided at the inlet moisture separator to avert any foreign matter from entering the moisture separator.

3.10.8.3 Discharge air filter. The compressed-air discharge shall be provided with a filter which shall have a nominal rating of 10 microns and an absolute rating of 25 microns conforming to MIL-F-5504. The filter element shall not impair the ability of the compressor to meet minimum performance requirements for at least 50 hours before filter change. The filter shall be in an accessible location so that removal and replacement of the filter element shall not require more than 10 minutes

3.10.9 Pressure relief valves. Adjustable, preset, spring-loaded pressue-relief valves shall be provided to protect the moisture separator, air receiver, and the individual stages of the air compressor. The sdjustment, once made, shall be securely locked and sealed to prevent unauthorized adjuctment. The last stage of the compressor pump may be protected by the use of a rupture disk. In addition, rupture disks shall be provided to protect the moistue separator and air receiver. The relief valves and rupture disks shall be capable of discharging 100 percent of the compressor capacity.

3.10.10 Receiver. A receiver shall be provided upstream of the dehydrator. The receiver shall withstand an operating pressure of 3,500 psi and, unless otherwise specified (see 6.2), shall have a volume capacity of not less than 1300 cubic inches. The receiver shall be equipped with a drain valve located at the lowest point of the receiver, a spring-loading-type pressure-relief valve set at 4,000 psi ±100 psi, and a disk-type pressure-relief valve set for a range of 4,400 to 4,800 psi. The construction shall conform to the Code for Unfired Pressure Vessels of the ASME Boiler and Pressure Vessel Code. The receiver shall bear the ASME stamp. Welding or other activation shall not be permitted after the receiver has been inspected. When a fiber glass receiver is utilized, it

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shall conform to MIL-T-25363 and to the requirements of this specification. A pressure vessel conforming to MIL-R-8573 is also acceptable provided it meets the requirements of this specification.

3.10.10.1 Unloader. A devices shall be provided unload the compressor automatically when the discharge pressure reaches 3,500 +100, -0 psi, an to load the compressor at a presssure between 3,000 psi and 3,500 psi.

3.10.10.2 Blowdown device. An automatic blowdown device shall be provided to discharge the air receiver and entire compression system when the engine stops. The device shall reduce the air receiver pressure to not more than 10 psi.

3.10.11 Moisture traps. Traps shall be mounted so as not to be affected by any extremes of temperature specified herein.

3.10. Sensitive elements. Sensitive elements such as fuel, cooling, and lubrication systems, storage batteries, pressure gages, drain valves, and connecting tubirq shall be so arranged within the enclosure as to prevent exposure to extremes of temperature.

3.10.13 Drain valves. Drain valves shall be grouped and mounted on a common member. Drain valve handles shall not extend beyond the exterior of the enclosure , shall have a surface temperature of less than 120°F, and shall be operable by personnel wearing heavy miten-type gloves. Drain valves shall be labeled as to the items drained(i.e., "RESERVOIR DRAIN" , "MOISTURE SEPARATOR DRAIN") and shall be provided with double ended arrows that are labeled at each end to indicate functional result (i.e., "OPEN" "CLOSE") .

3.10.14 Drive belts. Drive belts shall conform dimensionally to RMA Engineering Standards for Multiple V-Belt Drives and shall conform to the material requirements of MIL-B-11040 l A mans for adjusting belt tension shall be provided.

3.10.15 Hose assemblies. Hose for hydraulic, oil, and coolant lines, excluding radiator hose assemblies, shall conform to MIL-H-52471 and MIL-H-51471/2. Hose assemblies for pneumatic lines shall conform to MIL-H-38390 or MIL-H-26666, -6 or -4 size.

3.10.16 Servicing hose A 30-foot pneumatic hose assembly of the size specified in 3.10.15 shall be provided. The hose shall be provided with a quick-disconnect fitting for an operating pressure of 3,500 psi and mating with the terminal fitting of the high pressure servicing system mounted onthe control panel (see 3.15). The cupling shall be two-way shutoff and shall be a Hansen series 2 HK or equal. It shall be constructed ofcorrosion-resistant steel. An air chuck which shall fit valve assemblies conforming to MS28889 shall be provided on the servicing end of the pneumatic hose assembly. The hose shall be stowed in an accessible place inside the enclosure. The hose shall be secured in a manner to prevent fraying caused by vibration during preparation for use, during usage, and during stowage. Two servicing valves shall be provided; one located between the servicing line and the air chuck, and the other located between the dehydrator outlet and the servicing line to control the flow of air into the servicing line. A manually operated value for bleeding shall be provided to relieve the pressure in the servicing line-prior to disconnecting. A

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spring-loaded valve shall be provided between the bleed valve and the atmosphere to prevent dirt and moisture from entering the hose assembly when pressure is relieved.

3.11 Lubrication. All surfaces requiring lubrication shall be provided with a means of lubricating.

3.11.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIl-HDBK-113, chapter 2. When the specified of the lubricant selected includes a requirements for a Qualified Products* (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL (see 6.4).

3.11.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II or III. " Fittings shall be located in protected positions. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, type II. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconect fasteners may be removed to provide accessibility to lubrication fittings.

3.11.3 Pressure-release device. A pressure-release device shall be provided where the use of pressurelubrication equipment will damage grease seals or other parts.

3.11.4 Enclosures with internal reservoir of lubricant Enclousres, such as gear cases and transmission housings which contain a reservoir of lubricants for lubrication of the parts enclosed, shall be equipped with dipsticks, check plugs nut less than 1/2-inch pipe size, or sight gauges to determine the level of the lubricant. Each enclosure shall be equipped with a drain plug. Each enclosure shall reequipped with a means for filling the enclosure with lubricant. The drain plug shall be located so that removal of the drain plug will result in complete drainage of the lubricant f- the enclosure. Drain outlets shall be located to allow lubricant to drain to a suitable container when the compressor ins in its normal position. Integral tubes or troughs may be used to convey drained lubricants from the drain plug to the drain outlet. Accessibility to the drain plug, the filling means, and the lubricant-level checking device shall be obtained without the removal or adjustment of accessories or parts. Housing side panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to facilitate this accessibility.

3.11.5 Delivery lubrication service When delivered, all components, except those for which preservatives are specified in 5.1, shall be lubricated. Grease lubrication, excludig lubrication of sealed bearigns, shall be with grease conforming to MIL-G-10924 or other grease specifications listed in MIL-HDBK-113. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperatures range as specified herein. Each lubricated component shall be tagged ina conspicuous place to indicate the temperature range and grades of oils and greases used.

3.11.6 Oil dilution. Oil dilution for starting and running the compressorat low outside ambient temperature shall not be acceptable.

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3.12. Class 1, gasoline-engine-driven and class 3, diesel-engine-driven. For the purpose of selecting the engine, the compressor shall be classified as a class II end item in accordance with MIL-STD-1410.

3.12.1 Commercial engine. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, except as otherwise specified herein. The engine shall be 4-cycle and shall be furnished as a complete, self-contained, power package including hand throttle control, fail-safe governor, starting system and engine stop switch. The engine shall have a low oil shutdown type control and either a high cylinder head temperature or a high coolant temperature shutdown control. All controls shall be approved by the engine manufacturer and shall be set at the engine manufacturer's recommended limits. The engine shall be capable of operating under all conditions specified herein using fuels and lubricants specified herein and in MIL-STD-1410 without exceeding the engine manufacturer's recommended operating limits. The engine shall be furnished with a 12-volt cranking and battery charging system in accordance with MIL-STD-1410. The battery charging alternator shall have a capacity of not less than 18 ampere capacity. A flywheel type alternator is acceptable. A dry type air cleaner with a restriction indicator (to warn the operator when the air cleaner requires servicing) shall be furnished in accordance with MIL-STD-1410. Standard commercial batteries shall be furnished. The battery cables shall be colored black and labeled "-" for negative and colored red and labeled "+" for positive in accordance with MIL-STD-1472 (see 3.7.1). A pictorial battery connection diagram shall be provided that is readable when the batteries are being serviced.

3.12.2 Power and speed rating. The power and speed rating of the engine shall be such that operation of the compressor under any of the operational requirements and conditions specified herein, will not require horsepower in excess of the net continuous horsepower rating of the engine for the applicable governed speed.

3.12.3 Air induction system. A heavy duty air cleaner and restriction indicator, in accordance with MIL-STD-1410, 4.2.11 shall be furnished.

3.12.4 Fuels and fuel tank.

3.12.4.1 Engine fuel. For class 1 compressors, the gasoline engine furnished shall operate on gasoline conforming to ASTM D 439. For class 3 compressors, the diesel engine furnished shall operate on fuel conforming to both VV-F-800, grade DF-2 and MIL-T-83133, grade JP-8.

3.12.4.2 Fuel tank. A fuel tank with a capacity for not less than 8 hours of operation at rated load shall be provided. Construction and mounting shall be such that vibration, tightening of the mounting devices, or contact with other components of the engine or compressor will not cause leaks to develop. A filler neck assembly conforming to MS35644 shall be provided and shall incorporate a removable strainer, and have a chain secured filler cap and a fuel tank cap gasket. The filler cap and chain shall be attached to the fuel tank by means other than welding or brazing. The filler cap shall be a screw cap that will prevent fuel from leaking when the compressor is tilted 15 degrees from the horizontal in any plane. The tank shall be provided with an air vent and a drain. The fuel tank for class 1 compressors shall be labeled "MOGAS ONLY". The fuel tank for class 3 compressors shall be label "DIESEL OR JP-8 FUEL ONLY". The

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labeling letter height shall be 2 inches tall in accordance with MIL-STD-1472, 5.5.1.1 (see 3.7.1). A standpipe at least 1/4-inch above the bottom shall be incorporated in the fuel tank outlet to prevent the entry of dirt and foreign matter into the system. The tank and all lines shall be located or positioned to minimize fire hazard or damage resulting from spilling, overflow, or drainage. Provision shall be made for connection to an auxiliary fuel supply. The fuel tank shall be equipped with a fuel level measuring device.

3.12.5 Exhaust system. The back pressure of the exhaust system, including the muffler and the spark arrestor, shall not exceed the maximum recommended by the manufacturer at maximum horsepower output and speed under all operating conditions specified herein. The exhaust manifold, muffler, and spark arrestor shall be positioned so that exhaust gases shall be directed away from the operator's position, fuel tank, compressor air intake, and any other component or function that may be adversely affected by exhaust gases or heat and shall remain at least 4 inches away from the fuel tank under all belt adjustments. The exhaust system shall be protected against the entry of rain and shall include drainage provisions to prevent accumulation of water and condensed vapors.

3.12.6 Clutch. The clutch shall be accessible for adjustment, replacement of parts, and lubrication (unless bearings with sealed-in lubrication are used). Adjustment of the clutch shall not require special tools. The nominal rating of the clutch shall be not less than 150 percent of the maximum engine torque. The clutch operating lever shall be accessible to the operator and labeled as to function and shall be provided with double-ended arrows showing the direction of operation and labeled at each end to indicate the functional result (engage and disengage) in accordance with MIL-STD-1472, 5.4.3.2.1.3 (see 3.7.1).

3.13 Class 2, electric-motor-driven.

3.13.1 Electric motor. The electric motor shall be a totally enclosed, 3-phase, continuous duty induction motor conforming to NEMA MG-1, and as specified herein. The motor shall be 3-phase, 220/440-volt, 60-cycle, and shall be of a design which will assure low starting current and normal torque characteristics. The horsepower required by the compressors, when performing at rated capacity, shall not be more than the continuous horsepower rating, including the service factor, of the motor. All electrical equipment and wiring shall conform to the NFPA 70.

3.13.2 Starter switch. A pushbutton, on-off switch shall be provided.

3.13.3 Conversion items. All conversion items necessary to convert from one voltage to another shall be provided.

3.13.4 Conversion instructions. Complete instructions for converting from one voltage to another shall be permanently affixed in an accessible location. The conversion instructions shall be clearly readable and shall be applied to a plate conforming to 3.23.

3.14 Enclosure. A protective housing, constructed of steel of 0.048 inch nominal thickness or of aluminum of 0.064 inch nominal thickness, shall be furnished. A metal pocket for technical publications shall be provided on the inside of the enclosure. The housing shall completely enclose the sides, ends,

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bottom, and top of the compressor. The housing shall be provided with tight hinged doors and panels which hold tightly and securely when closed against extremely adverse and severe weather conditions. The enclosure shall prevent the entry of wind, water, dust, and snow. Drip channels or other similar facilities shall be provided to prevent entry of snow or rain. The enclosure shall not be welded to the frame and shall be capable of installation and removal without special tools. Provisions shall be made for the entry of cooling air into the interior of the compressor. The enclosure shall not be used to support heavy accessories, such as battery boxes and fuel tanks.

3.14.1 Doors, panels, and openings. Doors or panels shall be furnished with recessed latches operable by personnel wearing heavy arctic mittens. The doors or side panels shall be of a size and location to afford accessibility to the compressor unit, control equipment, and engine for routine service, adjustment, and maintenance. Doors over 18 inches in vertical height shall be hinged horizontally in two sections. Hinges shall be the continuous piano type with brass hinge pins. Doors shall be self supporting in the open position. Doors and panels shall be labeled to indicate the item(s) accessed through them. The roof of the enclosure shall be removable so that installed components can be removed or replaced vertically.

3.14.2 Storage rack. A storage rack shall be furnished and installed inside each compressor unit for wrapping or looping the air hose when not in use. The rack shall be installed in the compressor unit so that the hose will not come in contact with any component of the compressor subject to high temperature or abrasion.

3.14.3 Toolbox. A toolbox made of not less than No. 14 gage sheet steel shall be furnished with the lid edges formed to overlap the top edges of the box to prevent the entry of dirt, show, and rain. The lid shall be provided with a lockable, over center toggle type fastener in accordance with MIL-STD-1472, 5.9.10 (see 3.7.1). The toolbox shall be fastened to the compressor in a protected position. The toolbox shall be of sufficient size to house the operating tools, maintenance tools, and publications and shall incorporate a provision for drainage.

3.15 Control panel. A control panel shall be provided for the compressor with all compressor gauges, meters, valves, switches, and other operator controls as specified herein. The control panel shall be located within the enclosure, shall be shock-mounted, and shall be accessible through a watertight, hinged cover of sufficient size to expose all controls and instruments when open. Opening of the control panel cover shall not vent the engine and compressor unit compartment. Each instrument, meter and control device shall be identified by a nameplate or by legible lettering on its face. All gauges and meters shall be uniformly flush mounted. A schematic airflow diagram, operating instructions, and safety precautions shall be attached to the panel or control panel cover, so as to be visible to the operator when the control panel is in the open position. The control panel shall include the following equipment:

a. Oil pressure gauge (for engine having forced-feed lubrication above 15 psig).
b. Ammeter.
c. Fuel gauge, liquid level.
d. Engine hour meter.

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e. Starter and ignition switches.
f. Throttle and choke, or governor controls.
g. Temperature gauge, engine oil, indicating.
h. Pressure gauge, discharge air pressure (psig).
i. Pressure gauge, compressor lubricating oil pressure (psig) (if forced feed lubricated).

3.16 Lighting. Electric lighting for the control panel and for the compressor compartment shall be provided. Electric lights shall be provided for blackout operation of the control panel. Incandescent lamps shall be shielded so that the bare lamp is not visible from aerial or ground observation. Arrangement of the blackout lamps shall be such that all instruments, control knobs, and switches are identifiable and readable. The type to be used shall have an adjustable lens cover which will vary the light from off to full intensity.

3.17 Winterization. When specified (see 6.2), a winterization system shall be provided to maintain a temperature of at least 0 °F inside the enclosure at an outside ambient temperature of -50 °F when the power unit is not running.

a. The heater on the class 1 and class 3 compressors shall be complete with its fuel control and combustion system. The fuel for the heater shall be supplied from the fuel tank which supplies the engine. The heater shall conform (except for capacity) to the functional requirements of MIL-H-3177, capacity optional, or MIL-H-3199, type I, capacity optional and it shall be designed for a 12 volt system. The heat shall be distributed to the engine and compressor crankcases and the battery by means of ducts, blowers, and heat controls. Exhaust gases shall not be vented inside the enclosure.
b. On the class 2 compressor, an immersion-type heater shall be furnished. The heater shall maintain oil at operating temperatures during standby, and shall attain oil operating temperatures in 60 minutes after being cold soaked to -50 °F.

3.17.1 Winterized enclosure. A metal enclosure shall be provided to completely enclose the top, bottom, sides, and ends of the compressor assembly, to prevent the entry of blowing snow, rain, and dust, and to maintain temperature while heating and operating the compressor at -50 °F. Provisions shall be made for opening side panels or doors and manually adjustable louvers at the engine end to allow ventilation for the engine and compressor unit at 120 °F without excessive overheating as specified in MIL-STD-1300 for the engine, and exceed the temperature ratings as specified by the compressor manufacture for the compressor unit. Weatherstripping conforming to MIL-R-14328 shall be provided on all doors and panels. Weatherstripping shall have a protective wear coating, and both the weatherstripping and protective wear coating shall be flexible at -50 °F. The weatherstripping shall not interfere with normal operation. Insulation shall be furnished on the interior of the enclosure and shall be nonasbestos mastic type and not less than 3/32-inch thick.

3.17.2 Heater duct connection. A heater duct connection shall be provided on the enclosure for connection of a 12-inch-diameter duct from a heater conforming to MIL-H-4607, class 3. The duct connection and cover shall conform to MIL-H-7365, size A. The duct connection cover shall be securely fastened to the enclosure by a keeperchain or other means to prevent loss when removed. The

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heater duct connection shall not protrude beyond outer extremes of the enclosure more than necessary to facilitate connection of the heater duct.

3.17.2.1 Ducting. Provisions shall be made for adequate ventilation and circulation of air for the compressor, while preheating and operating under cold weather conditions. The heat from the heater shall flow through the compartment in such a manner as to warm the entire engine and compressor unit to permit starting of the compressor in not more than 30 minutes at -50 °F. Ducting or baffles shall be provided to divert heated air to the batteries and to all critical moving parts which may be restricted due to ice or congealed oil or grease. The heat shall be exhausted form the compartment through a manually operated louver in such a manner as to flow over the receiver, hose rack, and hose, warming these parts prior to starting.

3.18 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.5):

a. For winterized compressor units, one heater, engine and shelter, ground type H-1, NSN 4520-00-203-0224.

b. For type I compressors, one military, 3/4-ton, 2-wheel, M116 trailer chassis conforming to MS53028, to aid the contractor in designing the skid base for mounting on the M116 trailer chassis.

3.19 Type I, skid-mounted.

3.19.1 Dimensions. The overall outside dimensions of the type I compressor, including the skid, shall be not more than 72 inches long by 60 inches wide by 46 inches high.

3.19.2 Weight. The weight of the type I compressor, including the skid, shall be not more than 2,200 pounds.

3.19.3 Fenders. When specified (see 6.2), the contractor shall furnish two fender assemblies. Each fender assembly shall consist of a fender and splash guard conforming to SAE J321, SAE J682, and SAE J683, and necessary mounting brackets, which shall restrict water and other road-surface substances thrown from the wheels. Each fender shall support a minimum of 300 pounds resting on any 1-square-foot area without permanent deformation. The front, rear, and outer edges of the fenders shall be rolled or angled to prevent brush damage and reduce road splash. The splash guards shall be interchangeable with guards conforming to MS51331 and shall incorporate anti-sail characteristics. The splash guards shall be of fiber glass reinforced plastic, polyester, or solid polyvinyl. The minimum physical properties of the guards shall provide an ultimate strength of 3,000 psi. When fenders are specified, instructions for the installation of the fenders shall be furnished with the skid-mounted compressor.

3.19.4 Mobility. The compressor shall be transportable while mounted on the M116 trailer chassis, as shown on MS53028, for not less than 50 miles at speeds of 25 mph over relatively smooth paved roads without any damage, twisting or

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distortion, or instability due to incorrect mounting. Mounting instructions including type, strength, and number of fasteners to be used shall be provided for depot mounting.

3.19.5 Crankcase drainage. Provisions shall be made to carry the compressor and engine crankcase oil drain to the outside edge of the skid base.

3.20 Type II, wheel mounted. The type II compressor shall be wheel mounted using the mounting base of the compressor as the trailer frame and using pneumatic tires adequate for the applicable load but not less than size 6.00-9. The rim size and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of tires furnished. The tires shall be new and unused, and shall be not more than 12 months old on the date of shipment to the government. The weight of the type II compressor shall be not more than 2,450 pounds. The maximum dimensions of the type II compressor enclosure shall be 72 inches long by 60 inches wide by 46 inches high. The axle assembly shall be of the through-axle type. Axle spindles shall be heat treated and ground. The axle shall be equipped with tapered roller bearings having provisions for fine adjustment. The axle shall be perpendicular to and symmetrical about the longitudinal centerline. The installation of the axle mounts shall be in accordance with the recommendations of the axle manufacturer. The axles of the trailer shall be unsprung, and the front axle, shall be trunnion mounted to compensate for the unevenness of the roadway. The knuckle steer front axle shall have a cramping angle of not less than 30 degrees. The thread or track of the trailer shall permit the above specified cramping angle with a clearance with the trailer frame of not less than 2 inches.

3.20.1 Brakes. Two-wheel parking brakes shall be provided. The parking brakes shall be the automotive type with internal expanding shoe, correct backing plates, and drum. The brakes shall be applied by means of a single manual control (hand lever) not less than 10 inches and not greater than 18 inches in length from the pivot point and that will simultaneously operate both brakes to lock the wheels against rotation. The lever shall be installed near the towbar where it will not be damaged or interfere with trussing of the unit for shipping. Brake actuation forces shall not exceed 200 pound-inches.

3.20.2 Towbar. The wheel mounted compressor shall have a hinged metal towbar approximately 48 inches long. The towbar shall withstand the normal loads of turning and maneuvering without deformation. The towbar on wheel mounted compressors shall be equipped with a lunette ring conforming to MS51338 and two safety chains equipped with safety-type grabhooks. Each chain shall be composed of a 0.24-inch or larger open link chain of sufficient length to reach 3 feet beyond the lunette.

3.20.3 Towing pintle. On wheel mounted compressors a towing pintle conforming to MS51335-2 shall be provided on the rear of the trailer chassis at a height from 12 to 18 inches above the ground.

3.20.4 Ground clearance. The compressor chassis shall have a ground clearance of not less than 8 inches for the pintle heights specified herein and a clearance of not less than 2 inches when ascending or descending a 20-percent slope.

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3.20.5 Towing speed plate. The compressor shall be towable for 50 miles at any speed up to 25 mph over smooth paved surfaces and shall be furnished with a plate as specified in 3.23. The plate shall specify that the maximum towing speed shall be 25 mph over smooth paved surfaces. The plate shall be mounted so as to be conspicuous when the wheel-mounted compressor is being coupled for towing.

3.21 Transportability. The compressor shall be transportable by highway, rail, water, and air modes.

3.21.1 Highway transport. The skid mounted compressor or compressor-trailer combination shall be capable of being towed over paved roads, unimproved surfaces, and cross country terrain at the towing speeds described in 4.5.2.12.5. When the skid compressor is loaded on a 5-ton truck, it shall meet the highway legal limits of CONUS and NATO countries.

3.21.2 Rail transport. The compressor unit and all attached equipment, in the rail transport configuration must withstand transport by rail without damage. The compressor unit shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.5.2.12.1 without failure, damage, or permanent deformation.

3.21.3 Air transport. The compressor at maximum gross weight shall be roll-on/roll-off (RORO) transportable in an operational configuration by C-130 and C-141 aircraft without need of shoring. Criteria for air transports defined in MIL-STD-1791.

3.21.4 Lifting provision. The skid mounted compressor or the compressor trailer combination shall be equipped with permanently affixed provision conforming to MIL-STD-209, class 1, or class 3. The skid mounted compressor shall be considered as a MIL-STD-209, type IV equipment load and the compressor trailer combination shall be considered as a MIL-STD-209, type III equipment load. Instructions shall be stenciled above or near each provision on the compressor warning that the lifting provision shall not be used when the compressor is mounted on the trailer chassis. Lift points shall be labeled "LIFT HERE" in accordance with 3.7.1.

3.21.5 Tiedown provisions. The skid mounted compressor or the compressor-trailer combination shall have permanently affixed provisions that permit fastening the compressor to floor or deck of a transport medium. Tiedown provisions shall meet the requirement of MIL-STD-209, class 2, or class 3 and with MIL-STD-209 for air transport restraint criteria. Tiedown points shall be labeled "TIEDOWN HERE" in accordance with 3.7.1. Combination lift and tiedown points shall be labeled "LIFT/TIEDOWN HERE" in accordance with 3.7.1. The skid mounted compressor shall be considered as a MIL-STD-209, type IV equipment load and the compressor trailer combination shall be considered as a MIL-STD-209, type III equipment load

3.22 Identification marking. The compressor shall be identified in accordance with MIL-STD-130. The marking shall be applied to the compressor on identification plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a conspicuous, protected location.

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3.22.1 Shipping data plate. Each end item, or independent component assemblage thereof, shall have a shipping data plate attached as near as practicable to the name plate. The shipping data plate shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material and in addition, shall show the silhouette of the compressor in transport position indicating the center of gravity, the location and capacity of the lifting and the tiedown attachments. For type I, skid-mounted, compressors, the plate shall be attached by screws, bolts, or rivets marked with the following information:

a. Length - inches.
b. Height - inches.
c. Width - inches.
d. Cube - cubic feet.
e. Gross vehicle weight - pounds.
f. Center of gravity - location, inches.

For type II, trailer-mounted, compressors, the registration numbers and other markings shall be in accordance with MIL-STD-642. The gross weight shall be marked on each forward side of the compressor. The tire pressure shall be marked on the side of the trailer near the tires.

3.22.2 Stenciling. The gross weight and location of the center of gravity on each end item shall be stenciled on each unit in a manner easily discernable to personnel handling the equipment.

3.23 Instruction plates. Each compressor shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the compressor. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts or rivets in a conspicuous protected location.

3.24 Treatment and painting. The compressor, components, and parts painted in commercial practice shall be cleaned, treated and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 conforming to MIL-C-46168. When specified (see 6.2), the compressor shall be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns except the first article may be camouflage green 383. Components which will reach temperatures of 450 °F or more shall not be overcoated with MIL-C-46168 top coat.

3.24.1 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.24.2 Government furnished property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.7).

Item No.	Description	Quantity
1	Camouflage paint pattern	As required

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3.25 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference requirements shall be in accordance with MIL-STD-461, group II of class C3.

3.26 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 composition I or II as applicable, 7 percent salicylanilide (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.26.1 Composition. Composition II shall be used only in the case when local air pollution regulations in the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that composition II material complies with South Coast Air Quality Management District Rule 442 - Usage of Solvents.

3.27 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fits or performance.

3.28 Workmanship. All parts, components and assemblies of the compressor including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and weld parts shall be clean and free from sand, dirt, fins, pits, sprues, scales, or any damaging extraneous material. The compressor shall be free of any defect that could impair its operation and serviceability.

3.28.1 Fabrication. Steel and aluminum used in the fabrication of the compressors shall provide original quality surface finish and shall be free from kinks and sharp bends. The forming of the material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true, and sharp edges and burrs shall be removed. Flame-cutting may be employed instead of shearing and sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

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3.28.2 Welding. The surfaces of parts being welded shall free of scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation of failure when the parts connected by the welds are subjected to proof and service loading. Welding procedures shall be in accordance with one of these codes.

3.28.2.1 Welders and welding operators. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification tests as prescribed by one of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ASME Boiler and Pressure Vessel Code, Section IX Welding and Brazing Qualification.
AWS D1.1-Structural Welding Code - Steel.
AWS D1.2-Structural Welding Code-Aluminum.

Contractors who make only horizontal welds need not qualify welders or welding operators for "all position welding". Certification shall be made available for review upon request by the contracting officer or the contracting officer's representative. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. All test results shall be made available for review by the contracting officer or the contracting officer's representative upon request.

3.28.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Bolt holes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. Bolts shall not extend more than 5 threads outward from their associated nuts.

3.28.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.28.5 Machine work. Tolerances and gauges for metal fits shall conform to the limitation specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article compressor shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article compressor shall be tested as specified in 4.5.2.1 through 4.5.2.16 as applicable for the class and type of compressor unit to be furnished. Failure of any test shall be cause for rejection. A first article test log (see figure 1) shall be furnished. For class 3 compressor units, JP-8 shall be used for all first article tests.

4.4 Quality conformance inspection.

4.4.1 Examination. Each compressor shall be examined in accordance with 4.5.1. Presence of one or more defects shall be cause for rejection. For class 3 compressor units, JP-8 or DF-2 can be used for the quality conformance tests.

4.4.2 Tests.

4.4.2.1 Individual. Each compressor shall be tested as specified in 4.5.2.2 and 4.5.2.3. Failure of a test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The compressor shall be examined for the following defects:

101. Material not as specified (see 3.3).

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102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the compressor (see 3.4).
106. Environmental operation not as specified (see 3.5).
108. Safety devices or guards missing or not as specified (see 3.6).
109. Human factors engineering not as specified (see 3.7.1).
110. Safety standards not as specified as defined in ANSI B19.1 (see 3.6).
111. Rated capacity not as specified (see 3.8.1).
112. Requirements when tilted not as specified (see 3.8.2).
113. Endurance not as specified (see 3.8.3).
114. Maintainability not as specified (see 3.9).
115. Air compressor pump not as specified (see 3.10.1).
116. Cooling not as specified (see 3.10.2).
117. Pressure vessels not as specified (see 3.10.3).
118. Pressure tubing and fittings not as specified (see 3.10.4).
119. Valves not as specified (see 3.10.5).
120. Moisture separator not as specified (see 3.10.6).
121. Dehydrator not as specified (see 3.10.7).
122. Mounting for cartridge holder not as specified (see 3.10.7.1).
123. Operating instructions for cartridge holder not as specified (see 3.10.7.2).
124. Pressure control valve for cartridge holder not as specified (see 3.10.7.3).
125. Intake air filter not as specified (see 3.10.8.1).
126. Moisture separator for inlet air filter not as specified (see 3.10.8.2).
127. Discharge air filter not as specified (see 3.10.8.3).
128. Pressure relief valves not as specified (see 3.10.9).
129. Receiver not as specified (see 3.10.10).
130. Unloader not as specified (see 3.10.10.1).
131. Blowdown device not as specified (see 3.10.10.2).
132. Moisture traps not as specified (see 3.10.11).
133. Sensitive elements not as specified (see 3.10.12).
134. Drain valves not as specified (see 3.10.13).
135. V-belts and pulleys not as specified (see 3.10.14).
136. Hoses not as specified (see 3.10.15).
137. Service hose not as specified (see 3.10.16).
138. Lubrication not as specified (see 3.11).
139. Lubricants not as specified (see 3.11.1).
140. Lubrication fittings not as specified (see 3.11.2).
141. Lubrication pressure relief valve not as specified (see 3.11.3).
142. Enclosures with integral reservoir or lubricant not as specified (see 3.11.4).
143. Delivery lubrication not as specified (see 3.11.5).
144. Oil dilution not acceptable as specified (see 3.11.6).
145. Class 1 and class 3 engine-driven power unit not as specified (see 3.12).
146. Commercial engine not as specified (see 3.12.1).

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147. Power and speed rating not as specified (see 3.12.2).
148. Air induction system not as specified (see 3.12.3).
149. Fuel not as specified (see 3.12.4.1).
150. Fuel tank not as specified (see 3.12.4.2).
151. Exhaust system not as specified (see 3.12.5).
152. Clutch not as specified (see 3.12.6).
153. Class 2 electric motor power unit not as specified (see 3.13.1).
154. Starter switch not as specified (see 3.13.2).
155. Conversion items and instructions not as specified. (see 3.13.3 and 3.13.4).
156. Enclosure not as specified (see 3.14).
157. Doors, panels, and openings not as specified (see 3.14.1).
158. Storage rack not as specified (see 3.14.2).
159. Toolbox not as specified (see 3.14.3).
160. Control panel not as specified (see 3.15).
161. Lighting not as specified (see 3.16).
162. Winterization not as specified (see 3.17).
163. Winterized enclosure not as specified (see 3.17.1).
164. Winterization heating duct connections and ducting not as specified (see 3.17.2 and 3.17.2.1).
165. Dimensions and weight for skid-mounted not as specified (see 3.19.1 and 3.19.2).
166. Fenders for skid-mounted not as specified (see 3.19.3).
167. Mobility of skid-mounted not as specified (see 3.19.4).
168. Crankcase drainage for skid-mounted not as specified (see 3.19.5).
169. Type II, wheel-mounted compressor unit not as specified (see 3.20).
170. Brakes for wheel-mounted not as specified (see 3.20.1).
171. Towbar and towing pintle for wheel-mounted not as specified (see 3.20.2 and 3.20.3).
172. Ground clearance for wheel-mounted not as specified (see 3.20.4).
173. Towing speed plate for wheel-mounted not as specified (see 3.20.5).
174. Transportability by highway, rail, and air not as specified (see 3.21.1, 3.21.2, and 3.21.3).
175. Lifting and Tiedown provisions not as specified (see 3.21.4 and 3.21.5).
176. Identification marking, shipping data plate, and stenciling not as specified (see 3.22, 3.22.1, and 3.22.2).
177. Instruction plates not as specified (see 3.23).
178. Treating and painting not as specified (see 3.24).
179. Camouflage pattern data not provided as specified (see 3.24.1).
180. Electromagnetic interference not as specified (see 3.25).
181. Fungus and moisture resistance and composition not as specified (see 3.26 and 3.26.1).
182. Interchangeability not as specified (see 3.27).
183. Workmanship not as specified (see 3.28).
184. Assembly incomplete, components missing or not as specified (see 3.9 and 3.28).
185. Parts, special tools, and items to be furnished with compressor missing or not as specified (see 3.9).
186. Fabrication not as specified (see 3.28.1).
187. Welding not as specified (see 3.28.2).
188. Welders or welding operators not qualified as specified (see 3.28.2.1).
189. Bolted connections not as specified (see 3.28.3).

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190. Riveted connections not as specified (see 3.28.4).
191. Machine work not as specified (see 3.28.5).

4.5.2 Tests.

4.5.2.1 Test log. A first article test log covering the run-in, endurance, cycling, high temperature, low temperature, and tilted operation shall be established tabulating the appropriate test data in a similar manner to that shown in figure 1. High temperature shutoff, moisture separator, precipitation, lifting and tiedown, pulling, towing, and rail impact test results, as applicable, shall also be recorded in this log.

4.5.2.2 Test conditions. Only the run-in, capacity, and the endurance tests (see 4.5.2.3, 4.5.2.3.1, and 4.5.2.4) shall be performed with out shelter and at the climatic conditions existing at the place of test. The compressor shall operate as specified herein without maintenance other than the contractor's recommended normal schedule maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test. The compressor shall be lubricated with military oils and greases. Oils shall be those designated for use at the ambient temperature at the place of test. Safety and human factor considerations shall be evaluated throughout testing.

a. Unless otherwise specified herein, sequence of tests is optional with the tester.
b. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to start of testing shall be performed during the test.
c. Safety and human factor considerations shall be evaluated throughout testing.
d. The class 1 and class 3 compressor shall be started using the electrical cranking system without external assistance.

4.5.2.3 Run-in. The compressor shall be operated for a run-in period as determined by the contractor but not less than 8 hours of operation, utilizing all operator controls. During the final hour of this test, capacity shall be determined as specified in 4.5.2.3.1. Inability of the compressor to produce the capacity or meet the rated capacity specified in 3.8.1 shall constitute failure of this test.

4.5.2.3.1 Capacity. Capacity shall be determined in accordance with the Low Pressure Nozzle Method of the ASME Performance Code for Displacement Compressors, Vacuum Pumps, and Blowers, PTC-9. Capacity may also be calculated from the weight of air delivered in a given time with a constant back pressure of 3,500 psi, wherein the pressure vessel utilized for weight measurement shall be not less than 1000 cubic inches, and capacity shall be calculated at intake conditions based on 15 cfm at ASME standard of 14.7 psia, 68 °F and 36 percent relative humidity.

4.5.2.4 Endurance. Conduct the test without shelter and at the climatic conditions existing at the place of test. The compressor unit shall be coupled to its power unit and shall be operated for a total of 500 hours as follows:

a. 400 hours with the compressor initially delivering not less than 15 cfm of free air at 3,500 psi.

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b. 50 hours with the compressor cycling not less than 10 times per hour between the cut-in and cutout pressure of the unloader. The output shall be adjusted so that the pressure will alternately build up to 3,500 psi, the compressor unload, and the pressure drop to the cut-in pressure.
c. 50 hours with the compressor operating for alternate periods of 10 minutes loaded and 50 minutes unloaded.
d. For the 400 hours of operation specified in (a), data listed in 4.5.2.4.1 shall be recorded every 2 hours, and a capacity computation shall be made after every 40 hours of operation. A graph shall be prepared, plotting capacity and corresponding air receiver discharge temperature, taken and logged every 40 hours of operation.
e. At the end of each 50-hour cycling test specified in (b) and (c), remove the compressor from cycling operation and operate the compressor at rated capacity for 1 hour. After 1 hour operation, make a capacity computation.
f. Capacity shall be determined utilizing the formula as set forth in 4.5.2.3.1.
g. Endurance test will not be credited in less than 2-hour intervals of continuous operation. Intervals of endurance test time of less than 4 hours terminated by malfunction will not be credited. All test time accrued during other successfully completed tests specified herein shall be credited. Operate all functional controls, including valves, not less than 50 times during the test. All safety relief devices shall be caused to operate at least one time. Operation of safety devices may be demonstrated independently of compressor operation. Determine intervals of dehydrator cartridge change as specified in 3.10.7.2.
h. Readings shall be taken only after at least 1 hour of continuous operation. Record appropriate remarks, including a record of all stops and adjustments. The amount of lubricants consumed and fuel added during the tests shall be recorded as they are added to the compressor test system.

4.5.2.4.1 Test data.

a. Compressor crankshaft speed, in rpm.
b. Discharge pressure measured at dehydrator outlet, in psi.
c. Temperature of air at receiver outlet, in degrees Fahrenheit.
d. Intercooler and aftercooler inlet and outlet surface temperatures, in degrees Fahrenheit. Thermocouples shall be used and shall be located on the cooler inlet and outlet piping as close to the cooler as physically possible.
e. Intercooler and aftercooler pressures, in psi.
f. Free air delivery rate, in cfm.
g. Ambient air temperature and pressure, in degrees Fahrenheit and inches Hg.
h. Lubricating oil temperatures, in degrees Fahrenheit.
i. Average temperature of air inside enclosure, in degrees Fahrenheit.
j. Dew point of delivered air, in degrees Fahrenheit.

4.5.2.4.2 Final performance. Upon completion of 500 hours of testing, repeat the capacity test specified in 4.5.2.3.1.

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4.5.2.4.3 Failure criteria. Any of the following shall constitute failure of this test:

a. Failure of any component which cannot be corrected in accordance with the contractor's recommended schedule of maintenance.
b. Inability of the compressor to operate as specified in 3.8 and 3.10.
c. Inability of the valves to operate as specified in 3.10.5.
d. A fuel tank capacity less than specified in 3.12.4.2.
e. Inability of unloader to operate as specified in 3.10.10.1.
f. Inability of the blowdown device to operate as specified in 3.10.10.2.
g. Inability of the pressure relief valves to operate as specified in 3.10.9.
h. A decrease in capacity of the compressor after the final rated capacity of more than as specified in 3.8.1.

4.5.2.5 Operation in tilted position. The compressor shall be tilted in four different positions at an angle of 15 degrees from the horizontal. In each of the different positions, start and operate the compressor for 30 minutes delivering rated capacity at 3,500 psi. Then allow the compressor to stand for 4 hours in the tilted position most likely to adversely affect the lubrication or fuel systems, after which the compressor shall be started and operated for 30 minutes. Evidence of leaks; malfunction; faulty lubrication, cooling, or fuel supply and distribution; lubricant spill-over; or reduced performance shall constitute failure of this test.

4.5.2.6 Environmental.

4.5.2.6.1 Low temperature. For the winterized unit, expose the compressor to an ambient temperature of -50 °F for no less than 4 hours or until all temperatures have been stabilized for at least 1 hour. The winterization system (see 3.17) may be used for not more than 1 hour for heating prior to starting. For the non-winterized unit, expose the compressor to an ambient temperature of -25 °F for no less than 4 hours or until all temperatures have stabilized for at least 1 hour. Start the compressor unit and operate under conditions specified at full load for a minimum of 4 hours. While delivering air at 3,500 psi at a compressor crankshaft speed of not less than that used in establishing rated capacity in 3.8.1, record the data listed in 4.5.2.4.1 at 15 minute intervals and capacity at the end of each hour. Evidence of malfunction or damage to the compressor, or inability to start and operate as specified shall constitute failure of this test.

4.5.2.6.2 High temperature. Expose the compressor to an ambient temperature of 120 °F for 4 hours or until all temperatures have stabilized for at least 1 hour while at this ambient temperature, start the compressor and operate at full load for 8 hours, delivering air at 3,500 psi at a compressor crankshaft speed of not less than that used in establishing rated capacity in 3.8.1. Record the data listed in 4.5.2.4.1 at 30-minute intervals during operation of the compressor. Evidence of malfunction or damage to the compressor, or inability to start and operate as specified shall constitute failure of this test.

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4.5.2.7 Moisture separator. Operate the compressor for a minimum of 15 minutes at a temperature of not less than 120 °F and a relative humidity of not less than 85 percent. This test may be run in conjunction with the high-temperature test. Any condensed water or oil carryover from the separator into the dehydrator shall constitute failure of this test.

4.5.2.8 Rain. Subject the compressor, in operating condition with all panels necessary for operation open, to a simulated rainstorm by spraying fresh water at a minimum rate of 3 inches per hour directed simultaneously at all four sides and on the top at an angle of approximately 45 degrees from the vertical. The apparatus for producing the spray shall emit water in the form of droplets rather than a fine mist. Expose the compressor to the spray for 15 minutes while not running, and then start and operate the compressor for 15 minutes at no load while the spray continues. Water leakage into air cleaners and crankcases or water accumulation on other components which prevents operation shall constitute failure of this test.

4.5.2.9 High temperature shutoff. Activate the high cylinder-head temperature control device to stop the engine to determine conformance to 3.11.1. The device may be activated after complete or partial disassembly from the engine and by the application of an outside high temperature source to simulate high cylinder-head temperature.

4.5.2.10 Cycling. Start the compressor and allow the receiver pressure to build up until the pressure-regulation control operates. Then reduce the receiver pressure until the pressure-regulation control activates to increase the pressure. Vary the pressure as above until a total of 20 high-pressure actuations and 20 low-pressure actuations have been reached. Inability of the pressure-regulation control to control the pressure as specified shall constitute failure of this test.

4.5.2.11 Exhaust and fuel system.

4.5.2.11.1 Engine exhaust. Measure the engine exhaust pressure from a pressure tap approximately 2 inches from the exhaust manifold flange with the compressor operating at the maximum speed and the horsepower output specified herein. Exhaust pressure in excess of that specified in 3.12.5 shall constitute failure of this test.

4.5.2.11.2 Fuel tank capacity. Fill the fuel tank with fuel in accordance with 3.12.4.1. Start the compressor and operate at rated capacity and pressure until the engine ceases to operate due to lack of fuel. The elapsed time of operation shall be recorded. No refueling or auxiliary fuel supply shall be allowed. Elapsed operation time of less than as specified in 3.12.4.2 shall constitute failure of this test.

4.5.2.12 Transportability.

4.5.2.12.1 Rail impact test. The compressors and their associated equipment, in their transport configuration, shall be tested in accordance with MIL-STD-810. Nonconformance to 3.21.2 or if any component breaks, cracks, or any permanent deformation shall constitute a failure of the test.

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4.5.2.12.2 Tiedown provisions. The tiedown provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.21.5. Inability to meet the requirements of 3.21.5 shall constitute failure of this test.

4.5.2.12.3 Slinging provisions. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.21.4. Inability to meet the requirements of 3.21.4 shall constitute failure of this test.

4.5.2.12.4 Air transportability. The air transportability requirement for the C-130 and C-141 aircraft in accordance with MIL-STD-1791 (see 3.21.3) shall be certified. Inability to provide certification shall constitute failure for meeting this requirement.

4.5.2.12.5 Road test. Tow the type II, trailer mounted compressor unit, for a distance of 100 miles at maximum safe speeds. Twenty-five miles of this travel shall be on cross-country terrain (see 6.6.3) and 75 miles shall be on primary roads. Operation on smooth, paved surfaces shall include not less than 15 abrupt stops from a speed of 25 mph in the forward direction, not less than five abrupt stops from a speed of 2 mph in the reverse direction, and not less than 10 sharp turns to the left and 10 to the right at not less than 25 mph. Radii of the sharp turns shall be not greater than 150 feet. Displacement or damage to welded, bolted, or riveted connections or failure to operate as specified shall constitute failure of this test.

4.5.2.12.5.1 Brakes. The type II compressor shall be towed a short distance on a paved surface free of loose material with the two-wheel parking brakes applied. Inability of the parking brakes to hold either braked wheel from rolling during towing shall constitute failure of this test.

4.5.2.13 Shakedown. The type I compressor shall be mounted on the military M-116 trailer chassis and shall be tested as specified in 4.5.2.12.5. Damage, twisting or distortion, or instability due to incorrect mounting of the type I compressor and fender assembly when furnished (see 3.19.3) on the M-116 trailer chassis shall constitute failure of this test.

4.5.2.14 Electromagnetic interference. The compressor shall be tested in accordance with MIL-STD-462 to determine conformance to 3.25. The contractor shall furnish the contracting officer the first article test report for evaluation. Failure to meet the requirements of 3.25 shall constitute failure of this test.

4.5.2.15 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.16 Maintainability. Maintenance operations required during testing shall be accomplished in accordance with 3.9. Nonconformance to 3.9 shall constitute failure of this test. During testing, maintenance support elements such as manuals (when furnished as required by the contract), lubrication

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charts, repair parts, and special tools shall be evaluated. Inability of the compressor to be properly maintained utilizing the support elements shall constitute failure of this test and shall require revision of the document(s).

4.6 Inspection comparison. The Government may select compressors at any time during the contract production period and subject these compressors to the examination specified in 4.5.1 and tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article model compressors. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison compressor to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison compressor(s) and shall be cause for refusal by the Government to continue acceptance of production compressors until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on compressors previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all compressors accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging. The preservation, packing, and marking shall be examined and tested to determine compliance with the quality assurance provisions of MIL-C-3600. The preservation, packing and marking shall be examined to determine compliance with the requirements specified in section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A and B. Each compressor, accessories, repair parts, tools and publication shall be preserved in accordance with the applicable level A or B requirements of MIL-C-3600.

5.1.2 Level C. Each complete compressor and all items issued with the compressor shall be preserved to afford protection against mechanical and physical damage. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage shall be protected by wrapping, pack compartmentization, or cartonizing of the individual item. Liquid cooling systems shall be filled with a fresh clean solution of 50 percent water and 50 percent MIL-A-46153 heavy duty ethylene glycol antifreeze, and a tag attached indicating the temperature to which the

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cooling system can be subjected before damage. All lubrication reservoirs, sumps, etc., shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used, the temperature range of lubricant used and grade of lubricant used. A deprocessing guide shall be finished indicating in detail the preservation materials to be removed and any other servicing required prior to placing the compressor in operation.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A and B. Each compressor, preserved as specified in 5.1, shall be packed in accordance with the applicable level A or B requirements of MIL-C-3600.

5.2.2 Level C. Each complete compressor and all items issued with the compressor, preserved as specified in 5.1, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules; and as a minimum compressors shipped loose or mounted on skids in lieu of being boxed or crated shall be protected with a minimum 6 mil waterproof plastic shroud. The shroud shall be placed over the compressor and secured to the compressor base. Openings in skids shall be a minimum of 3 inches in height to provide access for forklift tines. Support legs of trailer mounted compressor shall be utilized for packing and storage only and shall not be used for support of the trailer during shipment.

5.3 Marking. In addition to any special or identification markings required by the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The compressors are intended to provide clean, dry, high-pressure air for missiles and flame throwers.

6.2. Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Time frame required for submission of first article model and number of first article compressors to be furnished (see 3.2).
e. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).

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f. When volume capacity of receiver other than as specified is required (see 3.10.10).
g. When winterization system is required (see 3.17 and 6.3.1).
h. When the property specified in 3.18 will not be loaned by the Government.
i. When tow fender assemblies are required for the type I compressor (see 3.19.3).
j. When electromagnetic interference requirements other than as specified in 3.25 are required.
k. Color required (see 3.24).
l. When camouflage pattern overcoatinging is not required (see 3.24).
m. When camouflage data will be provided (see 3.24.1 and 6.2.2).
n. Level of preservation and packing required (see 5.1 and 5.2).
o. Any special marking (see 5.3).

6.2.1 Data requirements. When this specification is used for an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

	Paragraph No.	Data requirements	Applicable DID No.
a.	3.24.1	Color photograph prints	DI-MISC-80192
b.	3.24.1	Camouflage line art data	DI-MISC-80176

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 Winterized first article. When winterized compressors are specified in addition to nonwinterized compressors, two first article models shall be furnished for examination and test. One of these first article models shall be a nonwinterized compressor and one shall be a winterized compressor.

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6.4 Lubricants. MIL-STD-838 prescribes the policy for using specification-type products wherever possible and provide specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22026-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.5 Government-loaned property. The contracting officer should arrange for loan of the Government property specified in 3.18 as required.

6.6 Definitions. For the purpose of this specification, the following definitions apply.

6.6.1 Pressure. All pressures referred to herein, unless otherwise specified as absolute or denoted by the symbol psia, shall be interpreted as psi gage.

6.6.2 Free air. Free air is defined as air at atmospheric condition surrounding the compressor, at the point where the compressor is installed, but not in the immediate vicinity where it may be affected by the heat of the compressor of engine cooling system.

6.6.3 Cross-country terrain. Cross-country terrain shall consist of generally level or improved open fields, broken ground, loose sand, or gravel road (see 4.5.2.12.5).

6.7 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.24.2.

6.8 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and special tools as necessary, and instructions on shipment of compressors. A suggested paragraph is as follows:

> "Shipment of compressors shall include repair parts, special tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

6.10 Classification of materials. The contracting officer shall require the contractor to provide certification of the equipment meeting ANSI B19.1 and the ASME standards.

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6.11 Subject term (key word) listing.

Compressor, air, diesel engine driven
Compressor, air, electric motor driven
Compressor, air, gasoline engine driven
Compressor, air, skid mounted
Compressor, air, trailer mounted

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME

Preparing activity:
Army - ME

Project 4310-A177

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DATA SHEET
MFG NAME_____ P/N_____ Record Time & Amount of 1. Engine Oil Added 2. Compressor Oil Added Sheet__of__

Test Tile:_____ Run No._____ Date_____

Item	SYM	UNITS	Readings											
			1	2	3	4	5	6	7	8	9	10	...	...
1. Time Start, Readings, Stop	R_t	HRS. MIN.	(IF STOP IS UNSCHEDULED GIVE REASON FOR STOP AND DOWN TIME)											
2. Temp. (Ambient) Dry Bulb	T_a	°F												
3. Temp. (Ambient) Wet Bulb	T_{wb}	°F												
4. Pressure (Ambient)	P_b	°F												
5. Temp., Engine Coolant	T_{e1}	°F												
6. Temp., Engine Lubricant	T_{e2}	°F												
7. Temp., Compressor Lubricant	T_L	°F												
8. Temp., Receiver Discharge	T_r	°F												
9. Pressure, Rec.	T_r	°F												
10. Pressure, Compressor Intake	Ps	°F												
11. Temp., Compressor Intake	T_1	°F												
12. Temp., Upstream of Nozzle	T_2	°F												
13. Pressure Differential Across Nozzle	dP	In.H_2O												
14. Compressor Speed	N	RPM												

FIGURE 1. Test log.

X-1904B

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-52037F(ME)	31 Jan 90

3. DOCUMENT TITLE
Compressor Unit, Reciprocating, Power-Driven, Air, 15 CFM, 3,500-PSI

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center, ATTN: STRBE-TSE, Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466, Telephone (703) 756-2340, AUTOVON 289-2340	

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